IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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(Click for GitHub repo)

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Outline

Executive Summary

- This project delivers an end-to-end analysis of SpaceX launch data and aims to predict Falcon 9 first stage landing success for Space Y, a competitor of SpaceX, using machine learning. Data was collected from SpaceX APIs and web scraping from Wikipedia, then meticulously cleaned and transformed using robust wrangling techniques, including creating a binary 'Class' variable for landing outcome.
- Exploratory Data Analysis (EDA): Through a combination of SQL queries and advanced visualizations—including interactive dashboards and spatial maps—the analysis revealed trends in payload dynamics, launch site performance, orbit type, and overall mission success rates. Visualizations showed relationships between flight number, payload mass, launch site, orbit type, and landing success. SQL queries provided insights into launch site distribution, payload mass, and landing outcomes.
- Interactive Visual Analytics: A Folium map was created to visualize launch sites and their proximities, and a Plotly Dash dashboard was developed to explore relationships between various features and launch outcomes.
- Predictive Analysis: Four classification models (Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors) were trained and evaluated, achieving an accuracy of approximately 83.3% on the test data. Predictive models were built to classify launch outcomes, highlighting critical factors that influence success.
- Conclusion: This comprehensive approach not only demonstrates applied data science methodologies but also provides actionable insights for strategic decision-making in aerospace operations. The models can predict Falcon 9 first stage landing success with reasonable accuracy, providing valuable insights for Space Y.

Introduction

- SpaceX has emerged as a transformative force in the aerospace industry, pioneering innovative launch strategies and technologies. This project leverages a comprehensive SpaceX launch dataset to dissect the factors influencing launch outcomes. By integrating data collection via API and web scraping, rigorous data wrangling, exploratory analysis through SQL queries and visualizations, and interactive analytics using maps and dashboards, we build an end-to-end data science pipeline.
- The project is structured to not only uncover key patterns in payload dynamics and launch success but also to deploy predictive models that forecast future outcomes. Through this detailed analysis, we aim to provide actionable insights that can enhance strategic decision-making and operational efficiency in aerospace endeavors. This initiative, part of the IBM Applied Data Science Capstone, demonstrates the practical application of advanced data science methodologies to solve complex, real-world problems.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Data was collected from SpaceX APIs and web scraping from Wikipedia
- Perform data wrangling
  - Data was meticulously cleaned and transformed using robust wrangling techniques, including creating a binary 'Class' variable for landing outcome
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Four classification models (Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors) were trained and evaluated, achieving an accuracy of approximately 83.3% on the test data.

Data Collection

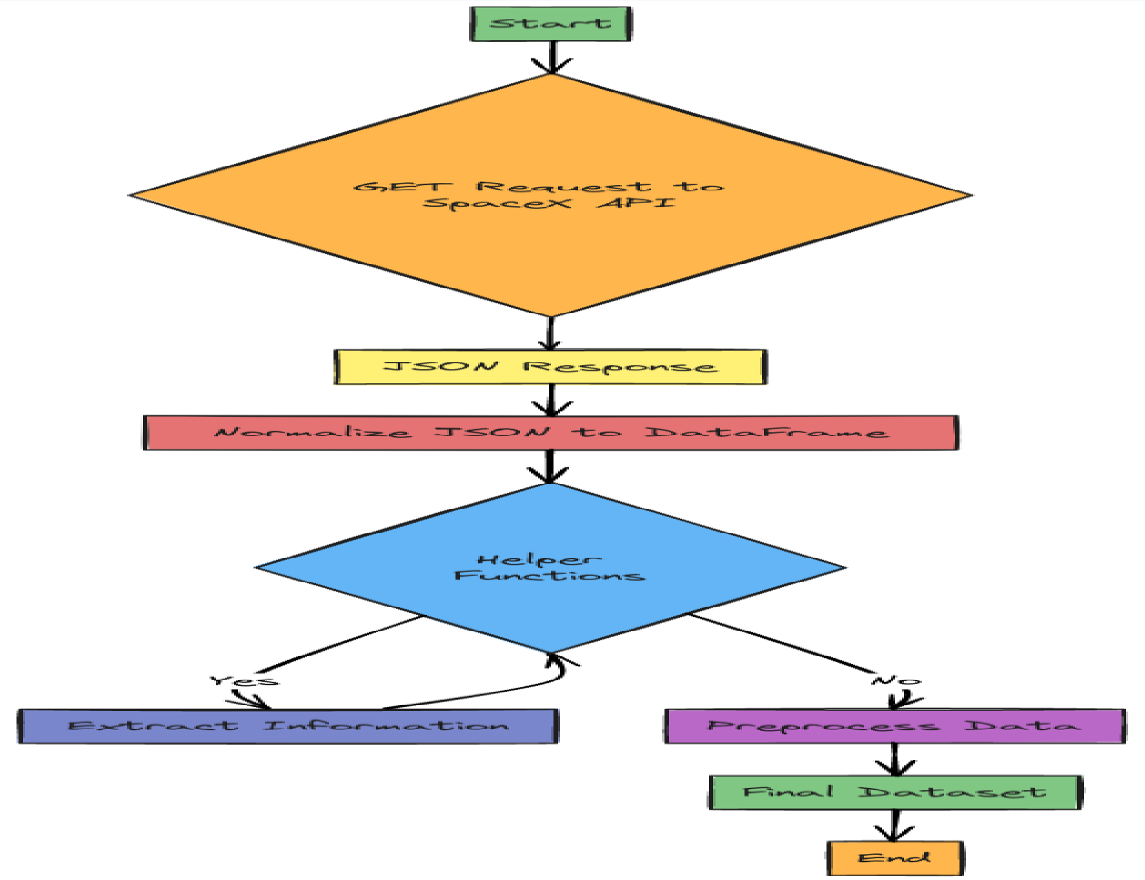
This project used a two-pronged data collection methodology to build a dataset for predicting Falcon 9 landing success and providing competitive intelligence:

- **SpaceX API**: RESTful API calls were made to retrieve detailed launch, payload, and outcome data, documented in Jupyter notebooks and visualized with a flowchart.
- **Web Scraping**: BeautifulSoup was used to extract supplementary data from Wikipedia, enriching the dataset with information not available via the API, also documented with a flowchart.

The data sources were the SpaceX API and Wikipedia, providing information on rocket launches, payload, and landing outcomes. The goal was to gather enough data to train a machine learning model and provide insights to a fictional competitor, Space Y, for pricing strategies. All methods are documented and accessible through GitHub repositories for transparency and reproducibility.

Data Collection – SpaceX API

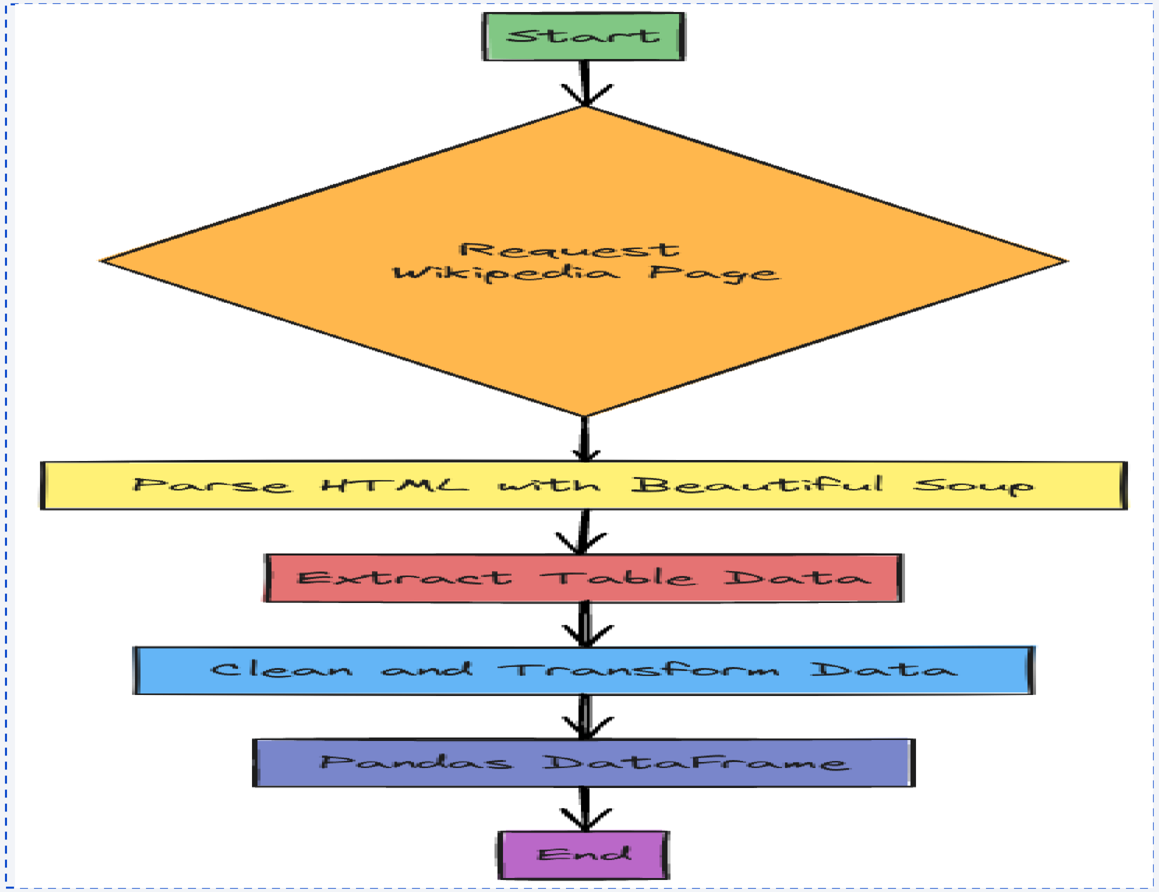
The SpaceX API, a RESTful service, provides real-time, structured data on SpaceX launches, including flight details, payloads, launch sites, and mission outcomes. This data was collected using GET requests, with the JSON responses transformed into Pandas DataFrames. Helper functions were used to extract specific information, and the data was preprocessed by filtering out Falcon 1 launches and handling missing values.



[GitHub Link to notebook]

Data Collection - Scraping

Web scraping was used to supplement SpaceX launch data by extracting contextual information from websites. Specifically, data on Falcon 9 launches was scraped from the "List of Falcon 9 and Falcon Heavy launches" Wikipedia page. BeautifulSoup was used to parse the HTML, extract data from tables, and transform it into a Pandas DataFrame after cleaning.



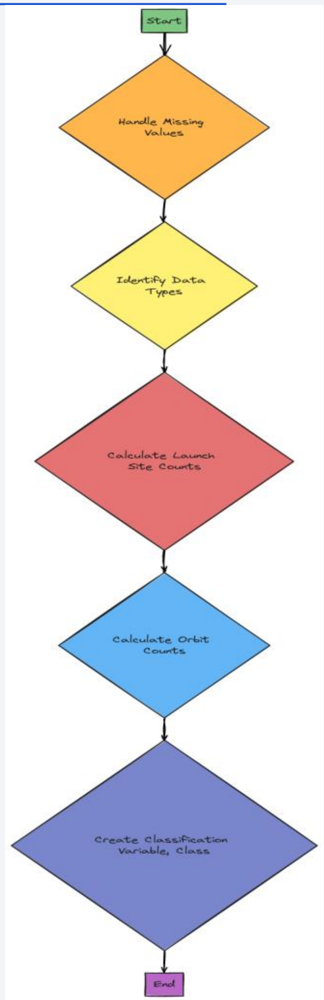
[GitHub Link to notebook]

Data Wrangling

Data wrangling transformed raw launch data into a clean, analysis-ready format through several key steps:

- Data Cleaning: Missing values were identified and handled, with "LandingPad" having the highest percentage of missing data. Inconsistencies and erroneous records were also addressed.
- Data Transformation: Data types were converted and values normalized for consistency.
- Data Integration: Data from the SpaceX API and web scraping were merged and launch site and orbit counts were calculated and integrated.
- Feature Engineering: A binary "Class" variable was created from the "Outcome" column, where 1 represents a successful landing and 0 a failed landing.

This process resulted in a prepared dataset with the "Class" variable as the target for machine learning models. Code snippets illustrate the key operations.



[GitHub Link to notebook]

EDA with Data Visualization

Exploratory Data Analysis (EDA) used various charts to visualize SpaceX launch data and identify factors influencing landing success:

Scatter Plots:

- Flight Number vs. Payload Mass: Showed increasing landing success with flight number and payload mass.
- Flight Number vs. Launch Site: Explored launch site impact.
- Payload Mass vs. Launch Site: Revealed payload mass limitations for certain launch sites.
- Flight Number vs. Orbit Type: Showed flight number and success relationships for different orbits.
- Payload Mass vs. Orbit Type: Showed payload mass and success relationships for different orbits.

Bar Charts:

- Success Rate of Each Orbit Type: Compared landing success across orbits.

Line Graphs:

- Launch Success Yearly Trend: Displayed the average launch success rate over time.

These visualizations revealed key relationships between flight number, payload mass, launch site, orbit type, and landing success, informing machine learning model development.

[GitHub Link to notebook]

EDA with SQL

SQL-based EDA was conducted on the SpaceX launch dataset using SQLite, extracting key insights through various queries:

Aggregate Data:
- SELECT SUM("PAYLOAD_MASS_KG_") AS total_payload_mass FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)'; (Total payload mass by NASA CRS)
- SELECT AVG("PAYLOAD_MASS_KG_") AS average_payload_mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1'; (Average payload mass by booster version F9 v1.1)

Filter and Group:
- SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE; (Unique launch sites)
- SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5; (Launch sites starting with 'CCA')
- SELECT "Landing_Outcome", COUNT(*) AS outcome_count FROM SPACEXTABLE GROUP BY "Landing_Outcome"; (Total successful and failure mission outcomes)

Identify Milestones:
- SELECT MIN("Date") AS first_successful_landing_date FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)'; (Date of first successful ground landing)
- SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000; (Boosters with successful drone ship landing)

Enhance Data Understanding:
- SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE); (Booster versions with maximum payload mass)
- SELECT substr("Date", 6, 2) AS month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE substr("Date", 0, 5) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)'; (Failure landing outcomes in 2015)
- SELECT "Landing_Outcome", COUNT(*) AS outcome_count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY outcome_count DESC; (Ranking of landing outcomes)

These queries provided insights into payload mass, launch sites, landing outcomes, and key milestones, complementing visualization-based EDA.

GitHub Link to notebook

Build an Interactive Map with Folium

This slide presents interactive maps created using Folium to visualize SpaceX launch sites, providing geographic context and insights into potential factors influencing landing success. These key map objects were employed:

- Markers: Pinpoint specific launch site locations.
- Circles: Represent a 1000m radius around each site, showing proximity.
- Pop-up Labels: Provide immediate launch site names.
- Mouse Position Plugin: Displays precise latitude and longitude coordinates.
- Color-coded Markers: Visually differentiate launch outcomes (success/failure).

These objects enable interactive exploration, spatial analysis, and the identification of geographic factors influencing launch success, such as distance to coastlines. Essentially, the map turns location data into an engaging tool to understand how geography affects SpaceX launch outcomes.

[GitHub Link to notebook]

Build a Dashboard with Plotly Dash

Overview: An interactive dashboard was developed using Plotly Dash to enable dynamic exploration of SpaceX launch data. It provides a user-friendly interface for visualizing launch success rates and payload mass correlations.

Key Components and Visualizations:

- **Dropdown Menu (Interactive Filter)**: Allows users to select specific launch sites or view data for all sites, filtering the data displayed in the pie chart and scatter plot.
- **Pie Chart (Dynamic Visualization)**: Displays the proportion of successful versus failed launches, updating in real-time based on the selected launch site.
- **Range Slider (Interactive Filter)**: Filters the data based on the selected payload mass range, enabling investigation of how payload mass affects launch outcomes.
- **Scatter Plot (Dynamic Visualization)**: Visualizes the relationship between payload mass and launch outcomes, with data points color-coded by booster version category, updating in real-time based on user selections.

Technical Implementation:

- **Callback Functions**: Dash callback functions connect user interactions to dynamic updates of the pie chart and scatter plot, ensuring responsive and data-driven visualizations.

Interactivity and Insights: The dashboard's interactivity allows for real-time exploration of data patterns, helping users identify:

- Launch sites with the highest success rates.
- Payload mass ranges that correlate with successful launches.
- Booster version categories associated with higher success rates.

GitHub Link to code

Predictive Analysis (Classification)

This section details the predictive analysis and machine learning process used to forecast Falcon 9 landing success. Key steps include:

Data Preparation:

- The dataset was split into training and testing sets.
- Features were selected and engineered.
- Data was standardized using StandardScaler.

Model Development:

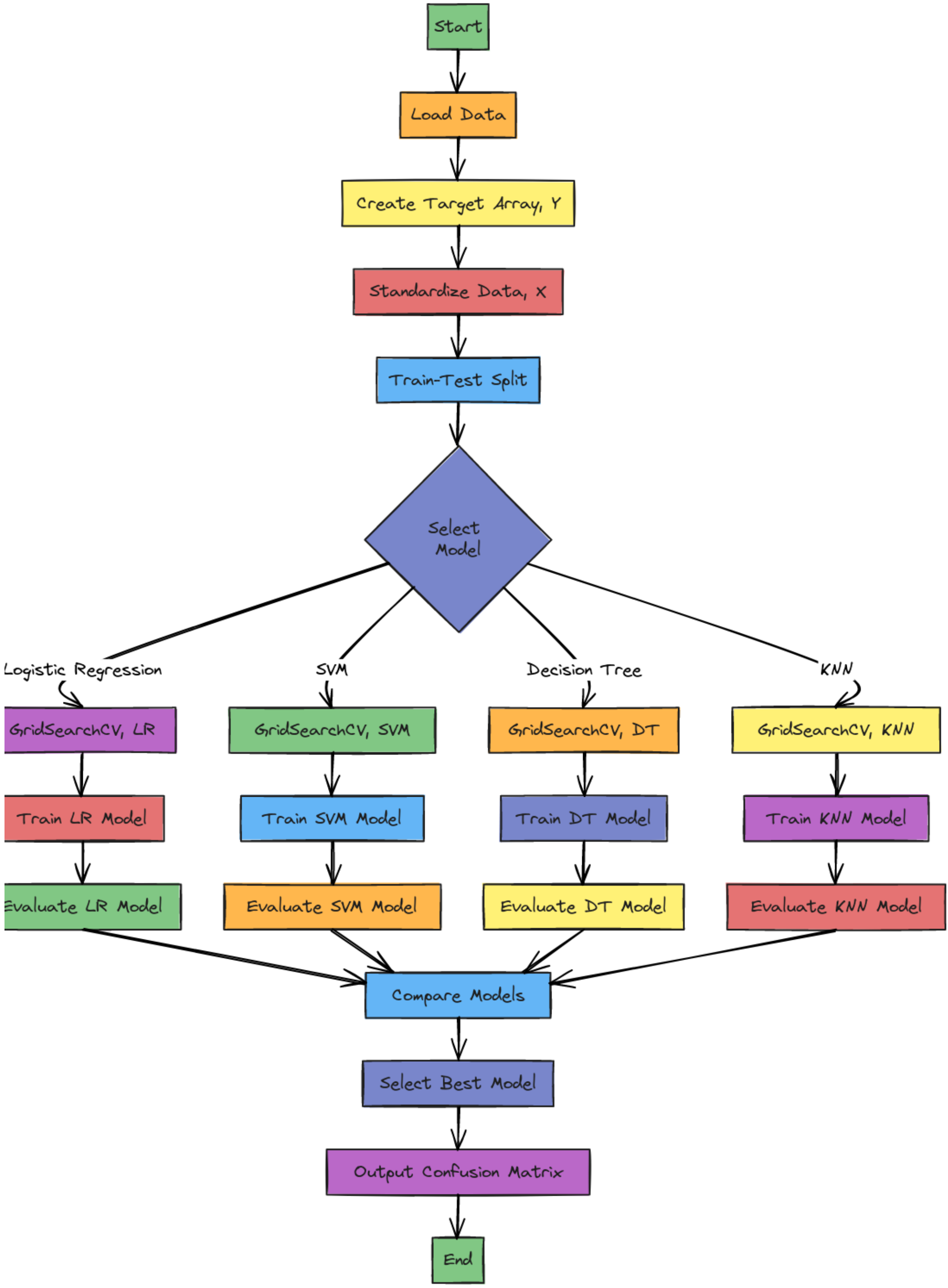
- Four classification models were trained: Logistic Regression, SVM, Decision Tree, and KNN.
- GridSearchCV was used for hyperparameter tuning.

Model Evaluation:

- Models were evaluated using accuracy, precision, and recall.
- Confusion matrices visualized performance.

Essentially, the process aimed to build and evaluate machine learning models for predicting landing success, with a focus on optimization and performance assessment.

[GitHub Link to notebook]



Results

This slide summarizes the project's key findings from various analysis methods:

- Visualizations: Scatter plots and bar charts showed trends like payload mass vs. success and launch site variations.
- SQL Analysis: Queries provided quantitative data on payload mass, success rates, and milestones.
- Interactive Mapping: Folium maps revealed geographic patterns and success/failure clusters.
- Dashboards: Interactive dashboards allowed for in-depth data exploration.
- Predictive Models: Classification models (Logistic Regression, SVM, Decision Tree, KNN) achieved approximately 83.3% accuracy in predicting landing success.

Key Findings:

- EDA showed correlations between flight number, payload mass, launch site, orbit type, and landing success.
- SQL queries detailed launch site distribution, payload mass, and landing outcomes.
- Folium maps visualized geographical factors.
- Plotly dash boards allowed for interactive data exploration.
- Machine learning models provided accurate predictions.

These results provide actionable insights for Space Y to understand SpaceX's operations and strategize accordingly.

Section 2

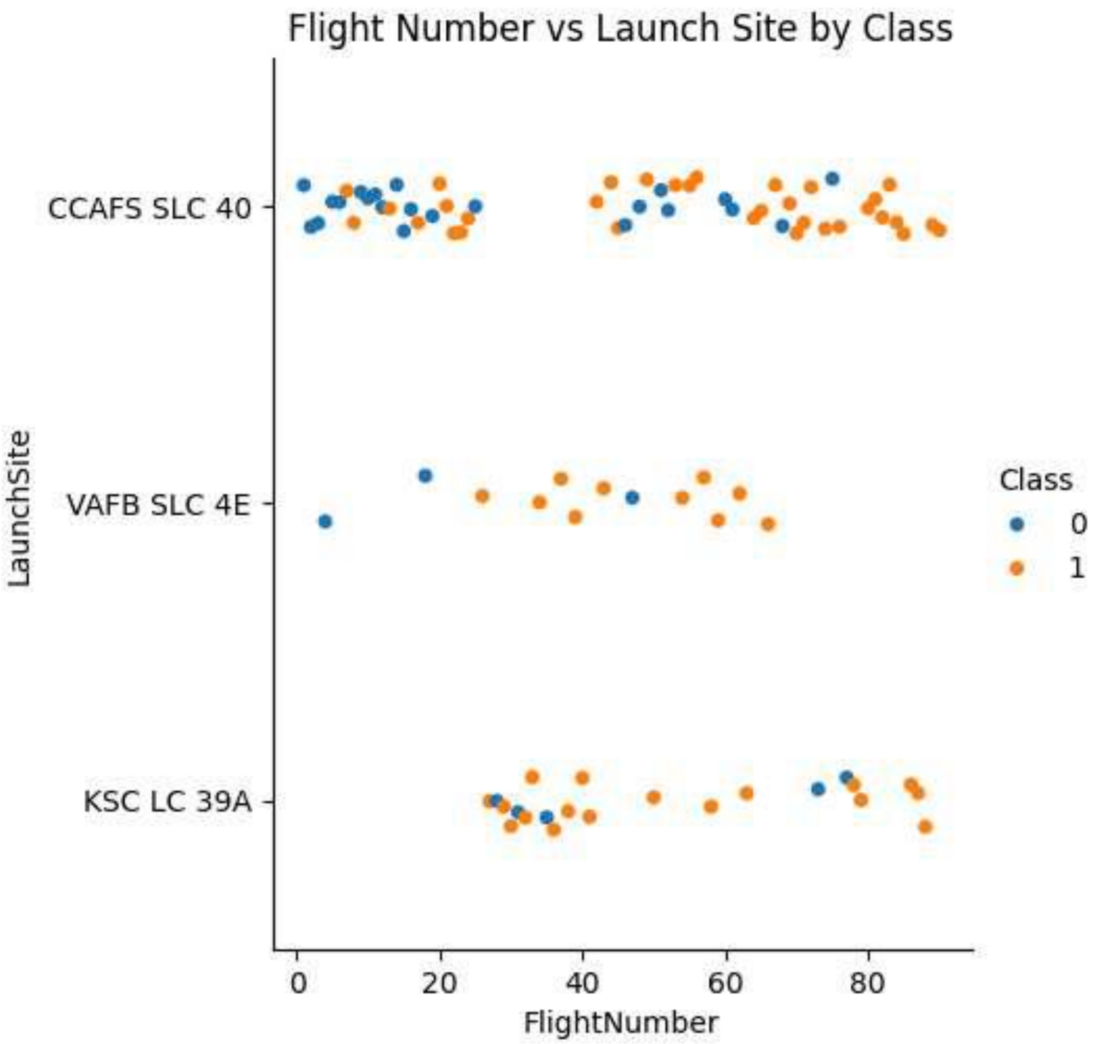
Insights drawn from EDA

Flight Number vs. Launch Site

This scatter plot analyzes SpaceX launch success by visualizing "Flight Number" vs. "Launch Site," color-coded by success (1) or failure (0). Key findings:

- **Site Clustering:** Each launch site forms a distinct cluster.
- **Flight Number Trend:** Success increases with flight number, indicating experience matters.
- **Site Variations:** Success rates differ by launch site.
- **KSC LC 39A Dominance:** Shows the highest success rate.
- **CCAFS SLC 40 Variability:** Exhibits mixed success/failure.
- **VAFB SLC 4E Limited Data:** Shows fewer launches, but notable successes.

Essentially, the plot demonstrates that both launch site and accumulated flight experience influence landing success, with KSC LC 39A being the most successful site.

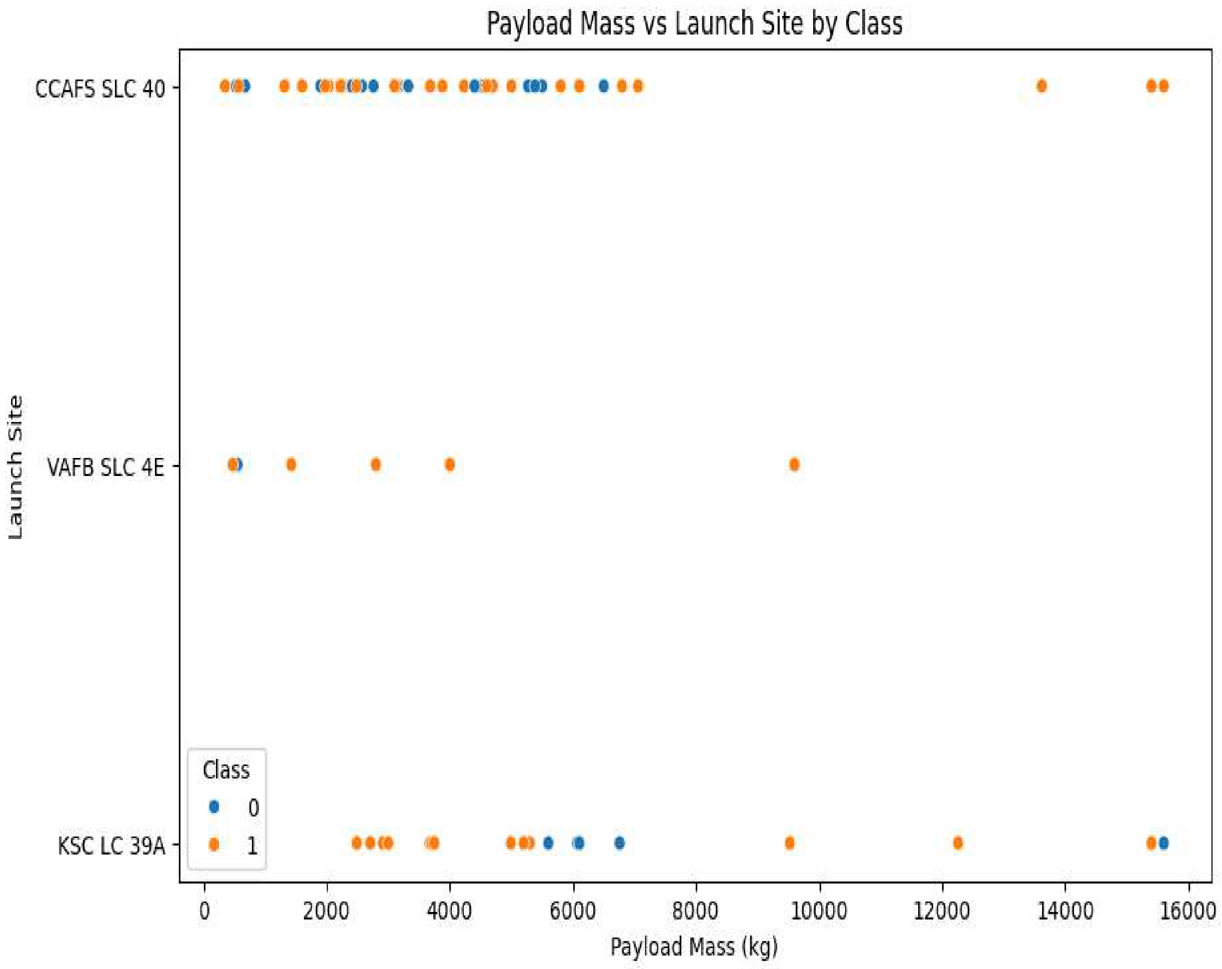


Payload vs. Launch Site

This scatter plot analyzes SpaceX launch success by visualizing "Payload Mass" vs. "Launch Site," color-coded by success (1) or failure (0). Key findings:

- **Site-Specific Payload Ranges:** Each launch site handles different payload mass ranges.
- **Outcome Distribution:** Successes and failures occur across all payload ranges.
- **Payload Mass Influence:** Payload mass appears to influence launch outcomes, potentially interacting with site-specific factors.
- **CCAFS SLC 40 Distribution:** Shows a wide range of payloads and mixed success, with no clear mass-success pattern.
- **VAFB SLC 4E Concentration:** Handles lighter payloads with a higher success rate.
- **KSC LC 39A Success with Variability:** Shows mixed success across a broad range of payloads.

Essentially, the plot indicates that both payload mass and launch site affect landing success, with VAFB SLC 4E showing a trend of higher success with lighter payloads.

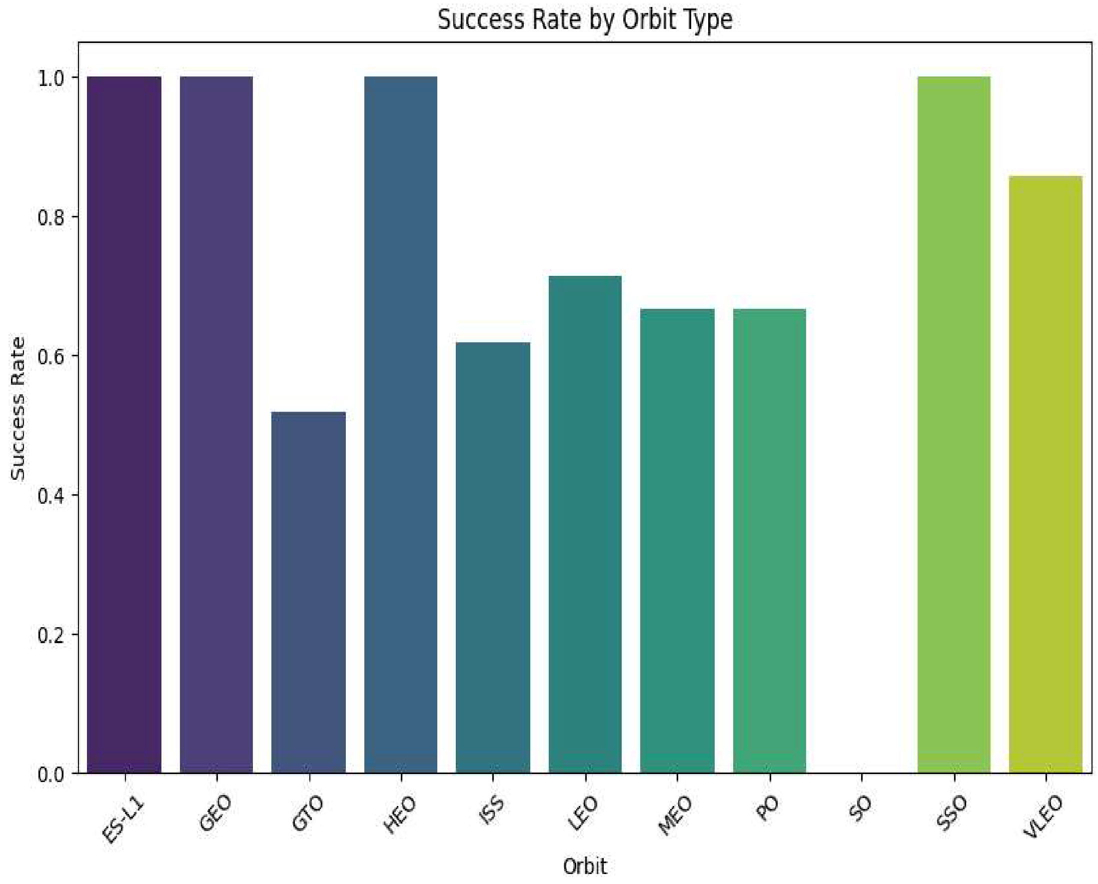


Success Rate vs. Orbit Type

This bar chart compares SpaceX launch success rates across different orbit types, revealing:

- **Varying Success:** Some orbits (SSO, VLEO, ES-L1, GEO, HEO) show high success, while others (GTO) have lower rates.
- **Orbit Influence:** Orbit selection significantly impacts landing success.
- **High Success:** ES-L1, GEO, HEO, and SSO have 100% success.
- **Moderate Success:** LEO, MEO, PO, VLEO, and ISS have moderate success (60-70%).
- **Low Success:** GTO has the lowest success rate (~50%).

In essence, the chart demonstrates that orbit type is a key factor in launch success, with certain orbits consistently yielding higher success rates than others.

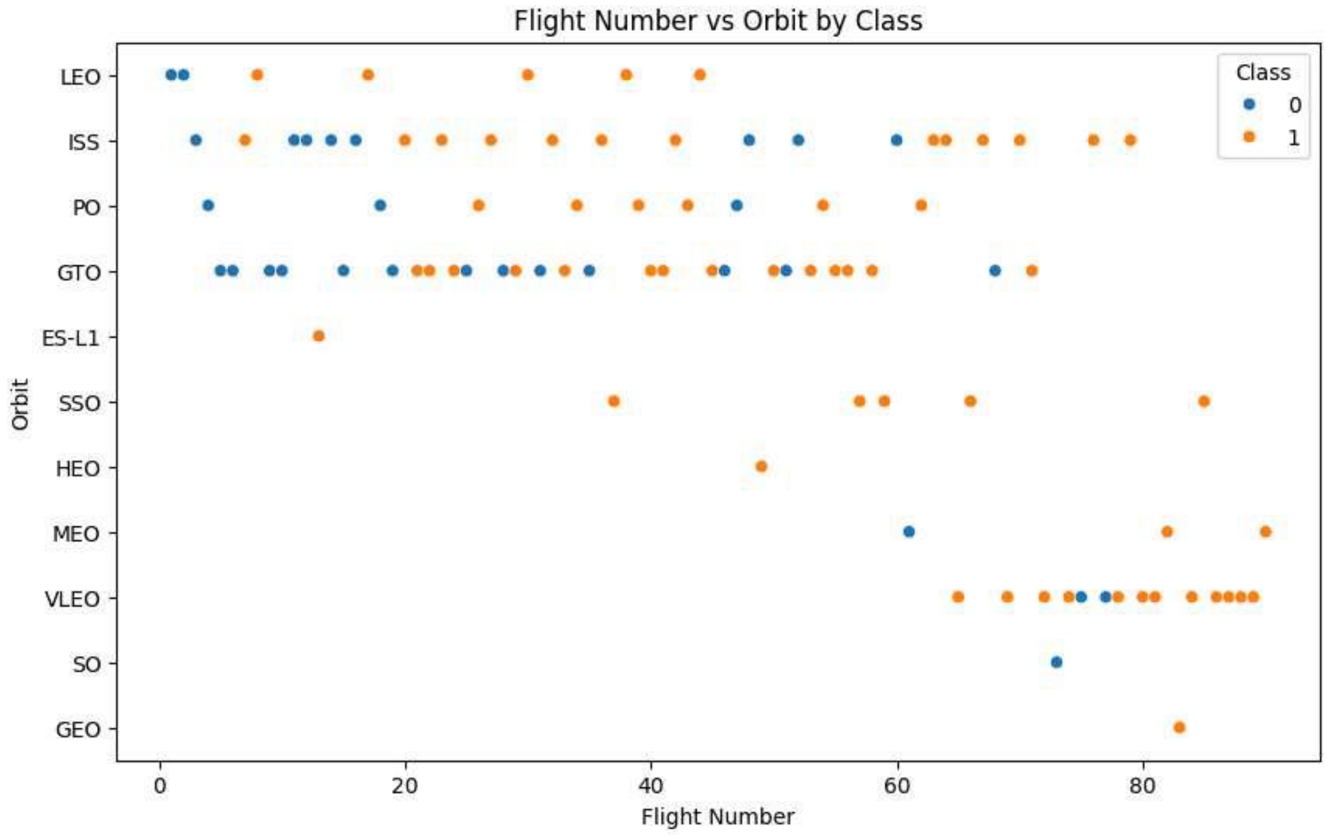


Flight Number vs. Orbit Type

This scatter plot analyzes SpaceX launch success by visualizing "Flight Number" vs. "Orbit," color-coded by success (1) or failure (0). Key takeaways:

- **Success Over Time:** Higher flight numbers correlate with increased success, indicating experience improves outcomes.
- **Orbit Variety:** Certain orbits appear at specific flight number ranges.
- **Complexity Differences:** Some orbits show mixed success/failure, reflecting varying mission complexity.
- **LEO and ISS Dominance:** These orbits show a higher concentration of successful landings.
- **GTO Challenges:** GTO exhibits mixed success/failure, suggesting inherent mission challenges.
- **Limited Data for Other Orbits:** Orbits like ES-L1, SSO, etc., have limited data, making conclusions difficult.

Essentially, the plot highlights that orbit type and flight number (experience) influence landing success, with LEO and ISS showing higher success rates, and GTO posing greater challenges.

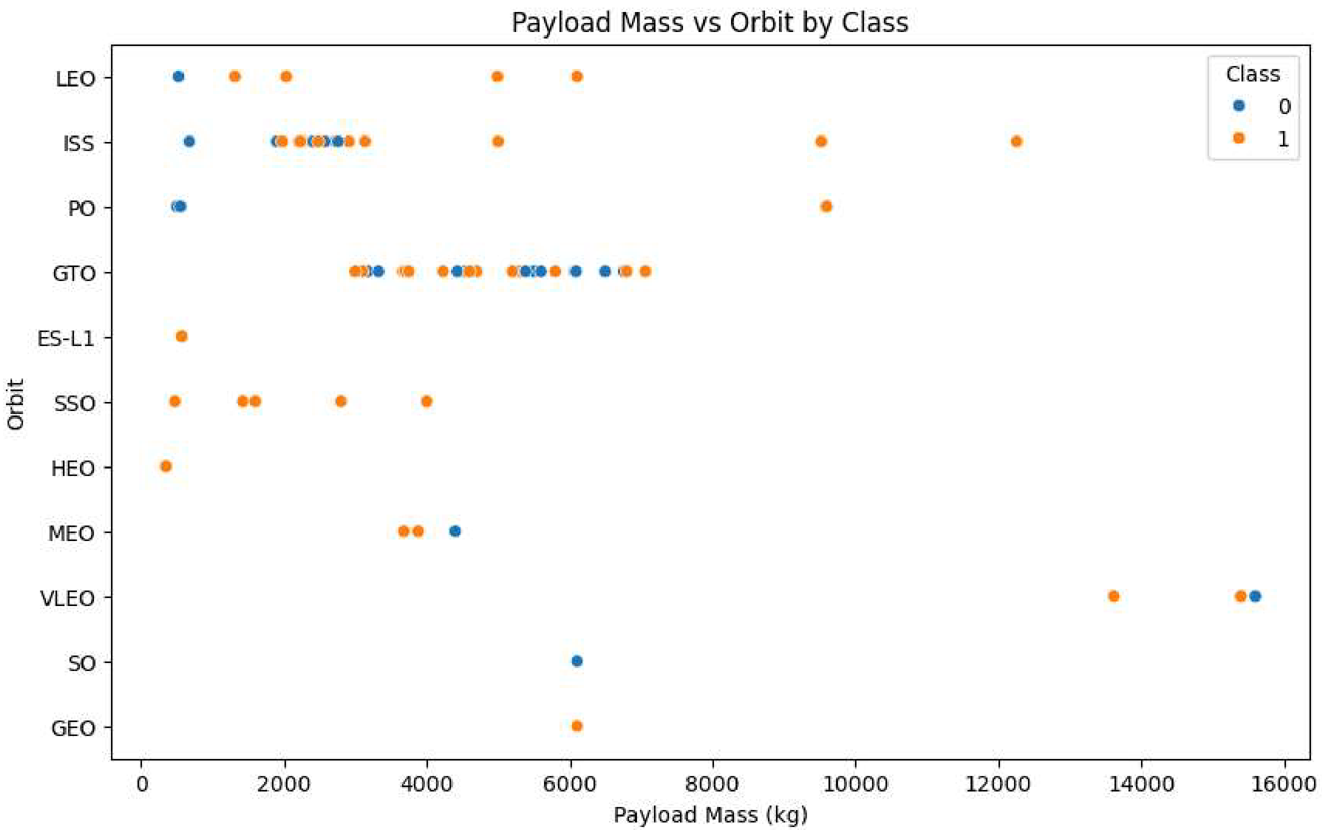


Payload vs. Orbit Type

This scatter plot analyzes SpaceX launch success by visualizing "Payload Mass" vs. "Orbit Type," color-coded by success (1) or failure (0). Key takeaways:

- **Orbit-Specific Payloads:** Different orbits handle varying payload mass ranges.
- **Outcome Variation:** Success and failure occur across all payload masses.
- **Orbit Influence:** Orbit and payload constraints impact launch outcomes.
- **LEO and ISS Success:** These orbits show high success across a wide range of payload masses.
- **GTO Challenges:** GTO exhibits mixed success, suggesting inherent challenges.
- **Varied Success in Other Orbits:** Other orbits show scattered success, indicating other influencing factors.
- **Payload Mass Impact:** Lower payloads tend to have more consistent success, except for LEO and ISS.

Essentially, the plot reveals that LEO and ISS are highly reliable, GTO is challenging, and other orbits' success is influenced by factors beyond payload mass.

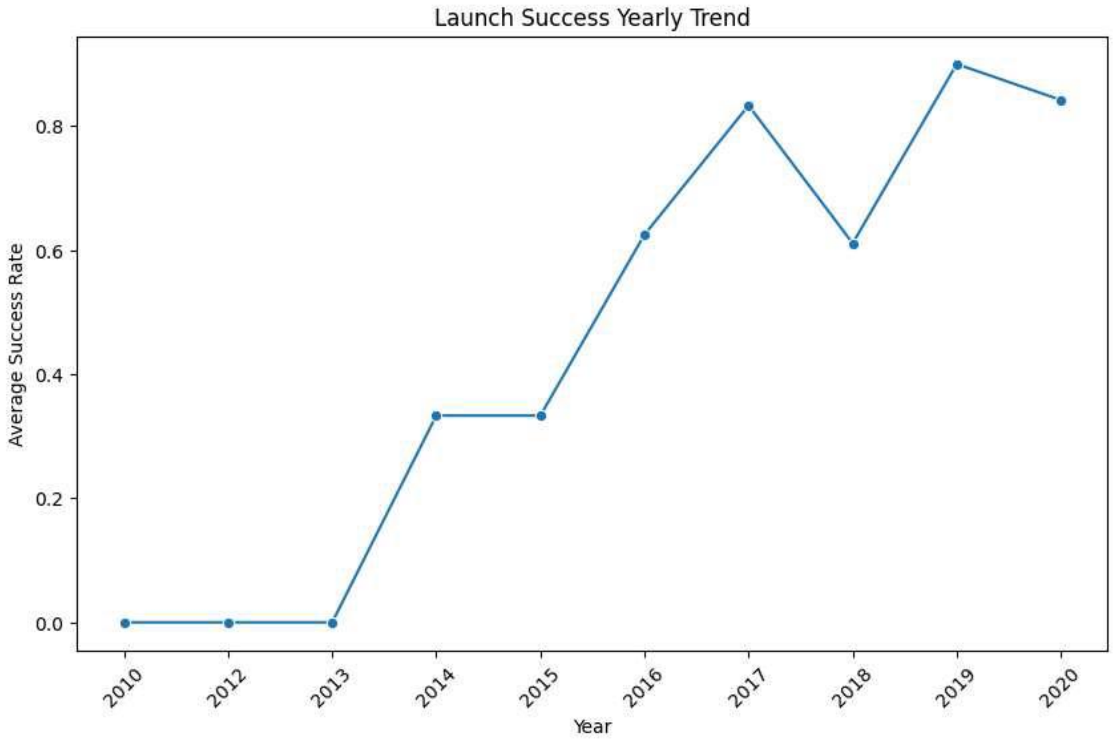


Launch Success Yearly Trend

This line chart depicts SpaceX's yearly average launch success rate from 2010 to 2020, revealing:

- **Upward Trend:** A clear and significant increase in success over time.
- **Operational Growth:** This reflects improvements in technology and procedures.
- **Reliability:** High success rates in later years indicate a mature program.
- **Early Challenges:** 2010-2013 showed no success.
- **Rapid Improvement:** 2014-2017 saw significant progress.
- **Peak Success:** 2018-2020 maintained high success with some fluctuations.

Essentially, the chart shows SpaceX's journey from early challenges to a highly reliable launch program, valuable information for competitive analysis.



All Launch Site Names

This SQL query identifies the unique launch sites in a dataset.

- Query: SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
- Result: The query returns four entries: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40.

Significance: This provides a basis for analyzing launch data by location, allowing for comparisons of success rates and other factors.

For Space Y: This helps Space Y understand where SpaceX operates and focus their analysis on these specific locations to inform their own strategies.

Task 1

Display the names of the unique launch sites

[11]:
```
query_task1 = '''
SELECT DISTINCT "Launch_Site"
FROM SPACEXTABLE;
'''

# Execute the query and display the res
unique_launch_sites = pd.read_sql_query
print("Unique Launch Sites:")
print(unique_launch_sites)
```

```
Unique Launch Sites:
    Launch_Site
0   CCAFS LC-40
1   VAFB SLC-4E
2    KSC LC-39A
3  CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

This SQL query retrieves the first 5 records from the "SPACEXTABLE" where the "Launch_Site" begins with "CCA".

- **Query**: SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
- **Result**: The query returns 5 records, all from "CCAFS LC-40", showing early SpaceX launch data.

Key Insights:

- Focuses on "CCAFS LC-40" data.
- Shows early SpaceX missions and landing challenges.
- Reveals information regarding payloads, dates, and mission outcomes.

For Space Y: Provides insight into SpaceX's early operations and landing development at "CCAFS LC-40", useful for competitive analysis.

Display 5 records where launch sites begin with the string 'CCA'

```
query_task2 = '''
SELECT *
FROM SPACEXTABLE
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5;
'''

# Execute the query and display the results
records_cca = pd.read_sql_query(query_task2, con)
print("Records where Launch_Site begins with 'CCA':")
records_cca
```

```
Records where Launch_Site begins with 'CCA':
```

	Date	Time (UTC)	Booster_Version	Launch_Site	
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon S
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Drag
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	
	2012-10-			CCAFS LC-	

Total Payload Mass

This SQL query calculates the total payload mass for NASA (CRS) missions.

- Query: SELECT SUM("PAYLOAD_MASS_KG_") AS total_payload_mass FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)’;
- Result: The query returns 45,596 kg as the total payload mass.

Key Insight: This shows the total amount of cargo SpaceX has delivered for NASA (CRS).

Significance: This provides insight into the scale of SpaceX's work with NASA.

For Space Y: This helps Space Y understand the payload requirements for NASA missions and inform their own competitive strategy.

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS

```python
query_task3 = '''
SELECT SUM("PAYLOAD_MASS__KG_") AS total_payload_mass
FROM SPACEXTABLE
WHERE "Customer" = 'NASA (CRS)';
'''


# Execute the query and display the result
total_payload_mass = pd.read_sql_query(query_task3, con)
print("Total Payload Mass for boosters launched by NASA (CRS):")
print(total_payload_mass)
```

```
Total Payload Mass for boosters launched by NASA (CRS):
   total_payload_mass
0               45596
```

Average Payload Mass by F9 v1.1

This SQL query calculates the average payload mass for the SpaceX booster version "F9 v1.1".

- Query: SELECT AVG("PAYLOAD_MASS_KG_") AS average_payload_mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
- Result: The query returns an average payload mass of 2928.4 kg.

Key Insight: This shows the typical payload capacity of the F9 v1.1 booster.

Significance: It provides data on the performance characteristics of this specific booster version.

For Space Y: This information allows Space Y to compare the F9 v1.1's capabilities with other boosters, understand SpaceX's technological development, and inform their own booster design and payload strategies.

Task 4

Display average payload mass carried by booster version F9 v1.1

```
query_task4 = '''
SELECT AVG("PAYLOAD_MASS__KG_") AS average_payload_mass
FROM SPACEXTABLE
WHERE "Booster_Version" = 'F9 v1.1';
'''

# Execute the query and display the result
average_payload_mass = pd.read_sql_query(query_task4, con)
print("Average Payload Mass for booster version F9 v1.1:")
print(average_payload_mass)
```

```
Average Payload Mass for booster version F9 v1.1:
   average_payload_mass
0                2928.4
```

First Successful Ground Landing Date

This SQL query identifies the date of SpaceX's first successful landing on a ground pad.

- Query: SELECT MIN("Date") AS first_successful_landing_date FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
- Result: The query returns "2015-12-22".

Key Insight: This marks a pivotal moment in SpaceX's landing technology development.

Significance: It signifies the achievement of controlled ground landings, essential for rocket reusability.

For Space Y: This helps Space Y understand SpaceX's development timeline and the point at which they achieved a crucial technological advantage, informing their own development strategies.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
query_task5 = '''
SELECT MIN("Date") AS first_successful_landing_date
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (ground pad)';
'''

# Execute the query and display the result
first_successful_landing_date = pd.read_sql_query(query_task5, con)
print("First successful landing outcome date on Ground Pad:")
print(first_successful_landing_date)
```

```
First successful landing outcome date on Ground Pad:
  first_successful_landing_date
0                    2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

This SQL query identifies specific SpaceX booster versions that successfully landed on a drone ship while carrying payloads between 4000 and 6000 kg.

- Query: SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000;
- Result: The query returns four booster versions: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2.

Key Insights:

- Identifies successful boosters within a specific payload range.
- Highlights the success of drone ship landings for these boosters.
- Shows the specific booster versions that have successfully preformed this task.

For Space Y:

- Allows Space Y to analyze successful booster designs and landing procedures.
- Helps assess the feasibility of drone ship landings for their operations.
- Provides data for competitive analysis and strategic planning.

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater tha

```
query_task6 = '''
SELECT DISTINCT "Booster_Version"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (drone ship)'
  AND "PAYLOAD_MASS__KG_" > 4000
  AND "PAYLOAD_MASS__KG_" < 6000;
'''

# Execute the query and display the results
successful_drone_ship_boosters = pd.read_sql_query(query_task6, con)
print("Boosters with success in drone ship and payload mass between 4000 and 6000:")
print(successful_drone_ship_boosters)
```

```
Boosters with success in drone ship and payload mass between 4000 and 6000:
  Booster_Version
0     F9 FT B1022
1     F9 FT B1026
2  F9 FT  B1021.2
3  F9 FT  B1031.2
```

Total Number of Successful and Failure Mission Outcomes

This SQL query counts the occurrences of each unique "Landing_Outcome" in the "SPACEXTABLE".

- Query: SELECT "Landing_Outcome", COUNT(*) AS outcome_count FROM SPACEXTABLE GROUP BY "Landing_Outcome";
- Result: The query returns a table showing each unique landing outcome and its count.

Key Insights:

- Reveals the frequency of successful and failed landings.
- Shows the variety of failure types (drone ship, parachute, etc.).
- Indicates the number of "No attempt" landings.
- Shows the counts of ground pad and drone ship successes.

For Space Y:

- Provides an overview of SpaceX's overall mission success rates.
- Allows for assessment of SpaceX's landing reliability.
- Informs strategic decisions regarding landing methods.

Task 7

List the total number of successful and failure mission outcomes

```
query_task7 = '''
SELECT "Landing_Outcome", COUNT(*) AS outcome_count
FROM SPACEXTABLE
GROUP BY "Landing_Outcome";
'''

# Execute the query and display the results
mission_outcomes = pd.read_sql_query(query_task7, con)
print("Total number of successful and failure mission outcomes:")
print(mission_outcomes)
```

```
Total number of successful and failure mission outcomes:
          Landing_Outcome  outcome_count
0       Controlled (ocean)              5
1                  Failure              3
2     Failure (drone ship)              5
3      Failure (parachute)              2
4               No attempt             21
5              No attempt               1
6   Precluded (drone ship)              1
7                  Success             38
8     Success (drone ship)             14
9      Success (ground pad)              9
10    Uncontrolled (ocean)              2
```

Boosters Carried Maximum Payload

This SQL query identifies the SpaceX booster versions that carried the maximum payload mass.

- Query: SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);
- Result: The query returns various "F9 B5" booster versions.

Key Insights:

- Highlights the "F9 B5" series' capability to carry the heaviest payloads.
- Demonstrates the high payload capacity of these boosters.
- Shows the technological advancement represented by the "F9 B5" series.

For Space Y:

- Provides data for analyzing the "F9 B5" series' performance.
- Helps understand SpaceX's technological lead in payload capacity
- .Informs strategic decisions regarding booster development.

Task 8

List the names of the booster_versions which have carried the maximum payload

```
query_task8 = '''
SELECT DISTINCT "Booster_Version"
FROM SPACEXTABLE
WHERE "PAYLOAD_MASS__KG_" = (
    SELECT MAX("PAYLOAD_MASS__KG_")
    FROM SPACEXTABLE
);
'''

# Execute the query and display the results
max_payload_booster_versions = pd.read_sql_query(query_task8, con)
print("Booster versions which have carried the maximum payload mass:")
print(max_payload_booster_versions)
```

```
Booster versions which have carried the maximum payload mass:
   Booster_Version
0    F9 B5 B1048.4
1    F9 B5 B1049.4
2    F9 B5 B1051.3
3    F9 B5 B1056.4
4    F9 B5 B1048.5
5    F9 B5 B1051.4
6    F9 B5 B1049.5
7   F9 B5 B1060.2
8   F9 B5 B1058.3
9    F9 B5 B1051.6
10   F9 B5 B1060.3
11  F9 B5 B1049.7
```

2015 Launch Records

This SQL query retrieves records of SpaceX drone ship landing failures specifically from 2015.

- Query: SELECT substr("Date", 6, 2) AS month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE substr("Date", 0, 5) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)';
- Result: The query returns two records, showing drone ship landing failures in January and April 2015

.Key Insights:

- Identifies specific drone ship landing failures in 2015.
- Shows the months in which these failures occurred.
- Provides details on the booster versions and launch sites involved.

For Space Y:

- Allows Space Y to analyze specific failure events and learn from SpaceX's experience.
- Highlights the challenges of early drone ship landings.
- Informs Space Y's own landing procedure development.

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to g

```
query_task9 = '''
SELECT substr("Date", 6, 2) AS month,
       "Landing_Outcome",
       "Booster_Version",
       "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 0, 5) = '2015'
  AND "Landing_Outcome" = 'Failure (drone ship)';
'''


# Execute the query and display the results
records_2015 = pd.read_sql_query(query_task9, con)
print("Records for months in year 2015 with failure landing outcomes on Drone Ship:")
print(records_2015)
```

```
Records for months in year 2015 with failure landing outcomes on Drone Ship:
  month       Landing_Outcome Booster_Version  Launch_Site
0    01  Failure (drone ship)   F9 v1.1 B1012  CCAFS LC-40
1    04  Failure (drone ship)   F9 v1.1 B1015  CCAFS LC-40
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

This SQL query ranks SpaceX landing outcomes between June 4, 2010, and March 20, 2017, by combining "Landing_Outcome" and "Landing_Type".

- Query: SELECT "Landing_Outcome" || ' (' || "Landing_Type" || ')' AS landing, COUNT(*) AS outcome_count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY landing ORDER BY outcome_count DESC;
- Result: The query returns a ranked list of landing outcomes, with "No attempt" being the most frequent.

Key Insights:

- Shows the frequency of different landing outcomes in the specified period.
- Highlights the early stage of drone ship and ground pad landing development.
- Reveals that many missions during this time did not involve landing attempts.

For Space Y:

- Provides insight into SpaceX's landing technology development timeline.
- Helps assess SpaceX's progress in drone ship and ground pad landings.
- Informs Space Y's own landing technology development strategies.

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (grour

```
query_task10 = '''
SELECT "Landing_Outcome" || ' (' || "Landing_Type" || ')' AS landing,
       COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY landing
ORDER BY outcome_count DESC;
'''

# Execute the query and display the results
landing_outcomes_ranking = pd.read_sql_query(query_task10, con)
print("Ranking of landing outcomes between 2010-06-04 and 2017-03-20:")
print(landing_outcomes_ranking)
```

```
Ranking of landing outcomes between 2010-06-04 and 2017-03-20:
                                   landing  outcome_count
0             No attempt (Landing_Type)             10
1    Success (drone ship) (Landing_Type)              5
2    Failure (drone ship) (Landing_Type)              5
3    Success (ground pad) (Landing_Type)              3
4      Controlled (ocean) (Landing_Type)              3
5    Uncontrolled (ocean) (Landing_Type)              2
6     Failure (parachute) (Landing_Type)              2
7   Precluded (drone ship) (Landing_Type)              1
```

Section 3

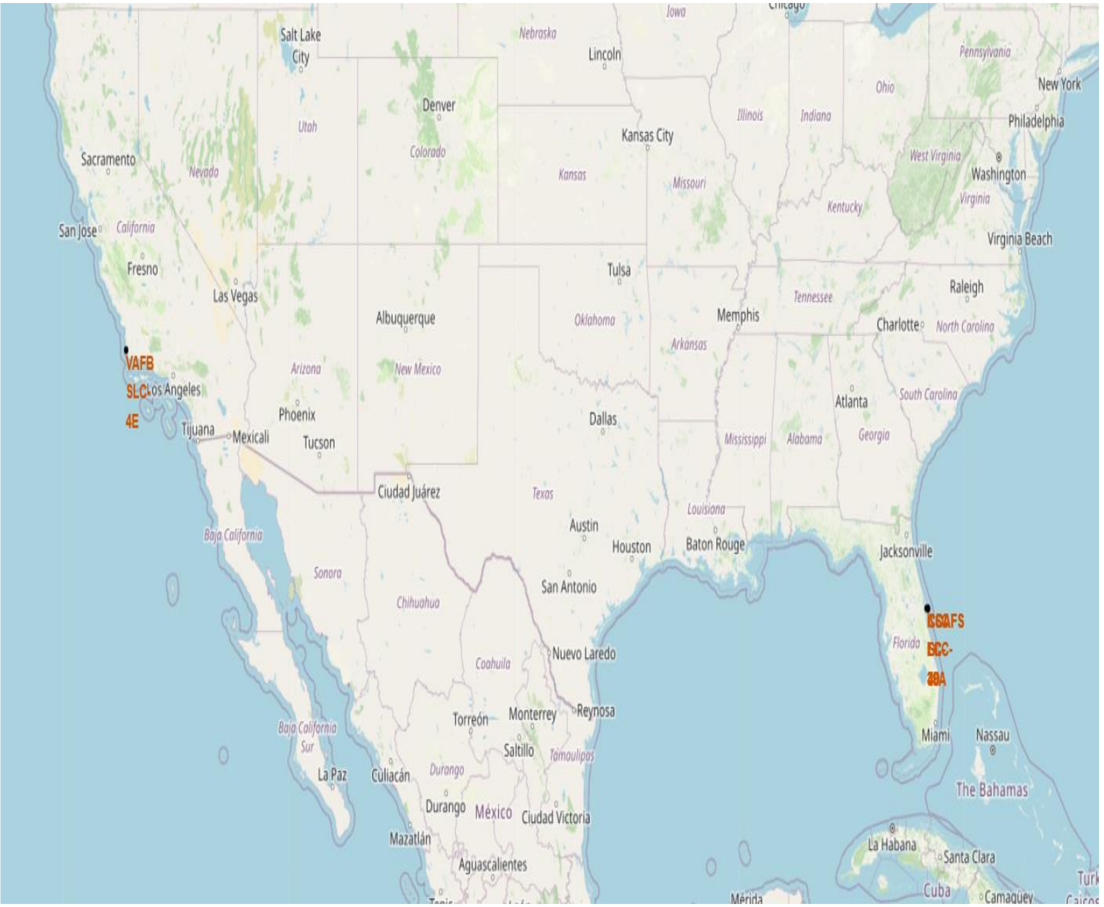
Launch Sites Proximities Analysis

SpaceX Launch Site Locations

This interactive Folium map shows major SpaceX launch sites across the US, allowing for **Launch Sites Proximities Analysis**. Key features include:

- **Launch Site Markers:** Clearly marks VAFB SLC-4E on the West Coast and KSC LC-39A/CCAFS SLC-40 on the East Coast.
- **Geographic Context:** Provides a US-wide view to understand relative distances.
- **Interactive Navigation:** Allows zooming and panning for detailed exploration.
- This spatial view highlights:
- **East Coast concentration:** KSC LC-39A and CCAFS SLC-40 are close together, suggesting potential logistical advantages and shared environmental factors.
- **West Coast isolation:** VAFB SLC-4E is geographically distinct, implying different launch considerations.
- **Distance and trajectory implications:** The separation between coasts likely influences launch paths and mission types.
- **Coastal locations:** All sites are coastal, highlighting the importance of this factor for space launches.

For Space Y, this map provides valuable information about SpaceX's launch site distribution and the strategic importance of location, which can inform their own launch infrastructure planning.

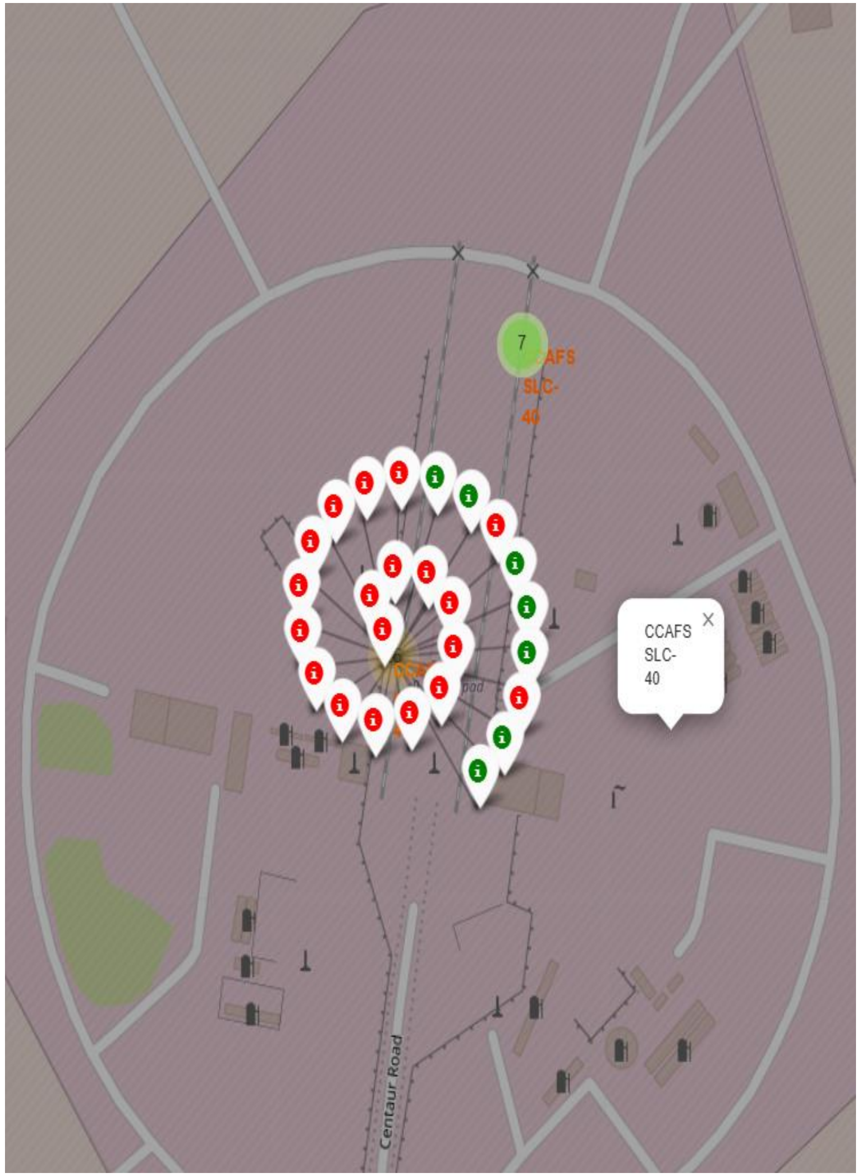


CCAFS SLC-40 Launch Outcomes

This Folium map provides a close-up view of the CCAFS SLC-40 launch site, using color-coded markers to show the outcome of each mission.

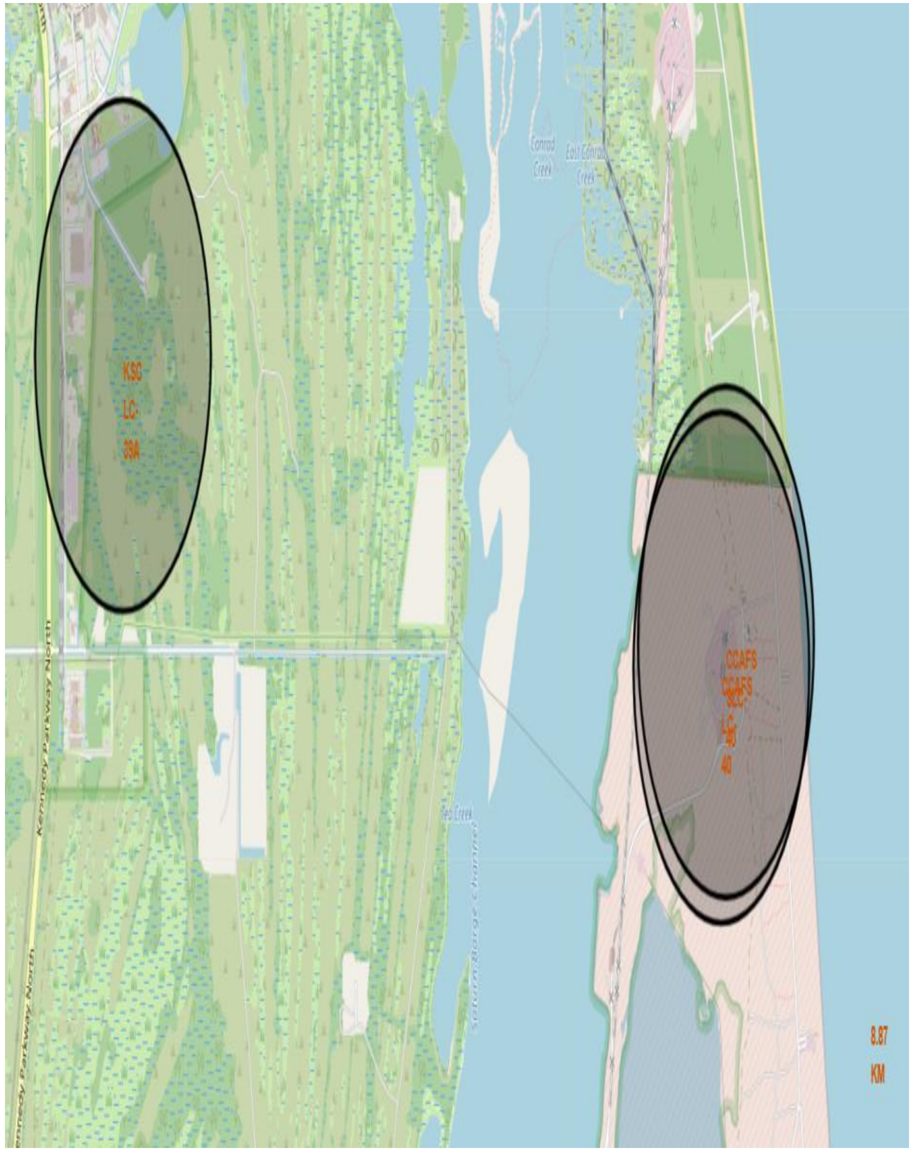
- Success/Failure Visualization: Green markers represent successful launches, and red markers indicate failures, allowing for a quick assessment of the launch history and success rate at this site.
- Spatial Patterns: The close proximity of markers enables the identification of any patterns or clusters of successful or failed launches.
- Site Layout: The map also shows the launch site's layout, including roads and other infrastructure.

This detailed view helps analyze launch outcomes at a specific site, providing insights into potential factors that may influence success or failure.



Proximity of Launch Sites to Coastline

- This interactive Folium map shows the proximity of SpaceX launch sites to the coastline.
  - Distance Visualization: It highlights the approximate 8.87 km distance between the launch sites (KSC LC 39A and CCAFS SLC 40) and the nearest coastline.
  - Safety and Logistics: This information is crucial for evaluating safety zones, environmental impacts, and the logistics of launch and recovery operations.
  - Coastal Emphasis: The map visually reinforces the fact that launch sites are strategically located near the coast for safety reasons, primarily to allow for launches over water.
  - Geographic Context: The map also provides a broader view of the surrounding landscape, giving context to the launch sites' location.
- This visualization helps understand the spatial relationship between launch sites and the coast, a critical factor in launch planning and safety.



Section 4

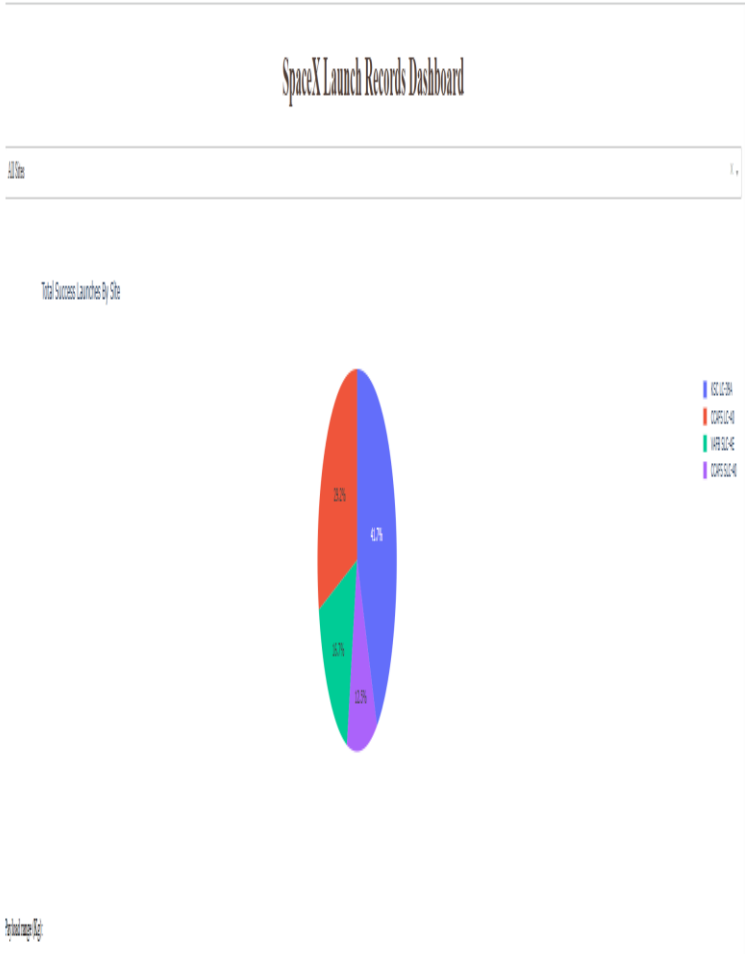
Build a Dashboard with Plotly Dash

SpaceX Launch Success by Site

This pie chart shows the distribution of successful launches across all SpaceX launch sites.

- Site Comparison: Each slice represents a launch site, with the size indicating its proportion of successful launches.
- Dominant Site: The largest slice highlights the most successful launch site.
- Relative Performance: Comparing slice sizes reveals the relative success rates of different sites.
- Clear Legend: The color-coded legend makes it easy to identify each site.

This visualization provides a quick overview of the success distribution across SpaceX launch sites, emphasizing the most productive locations.

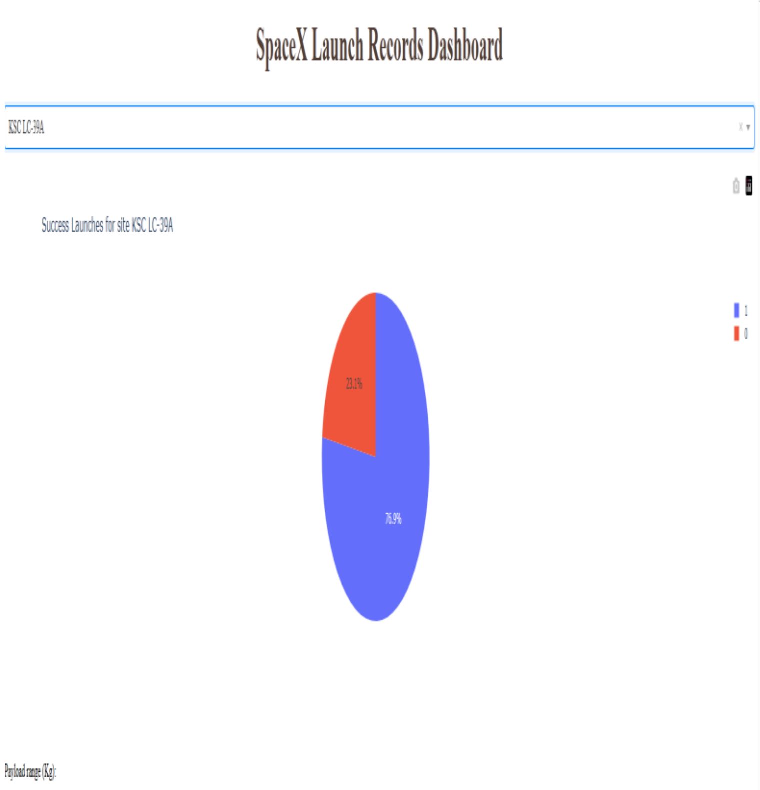


Highest Success Ratio – KSC LC-39A

This pie chart visualizes the launch success rate specifically for the KSC LC-39A launch site.

- Success/Failure Proportion: The blue segment represents successful launches (76.8%), and the red segment represents failures (23.2%), highlighting the site's high success ratio.
- KSC LC-39A Focus: This chart isolates KSC LC-39A, allowing for a focused analysis of its performance compared to the previous chart that showed all sites.
- Reliability: The dominant blue segment emphasizes KSC LC-39A's reliability and significant contribution to SpaceX's overall mission success.

This visualization effectively shows KSC LC-39A as a launch site with a high probability of success, making it a potentially favorable location for future missions.



Payload Range Filter and Launch Outcome

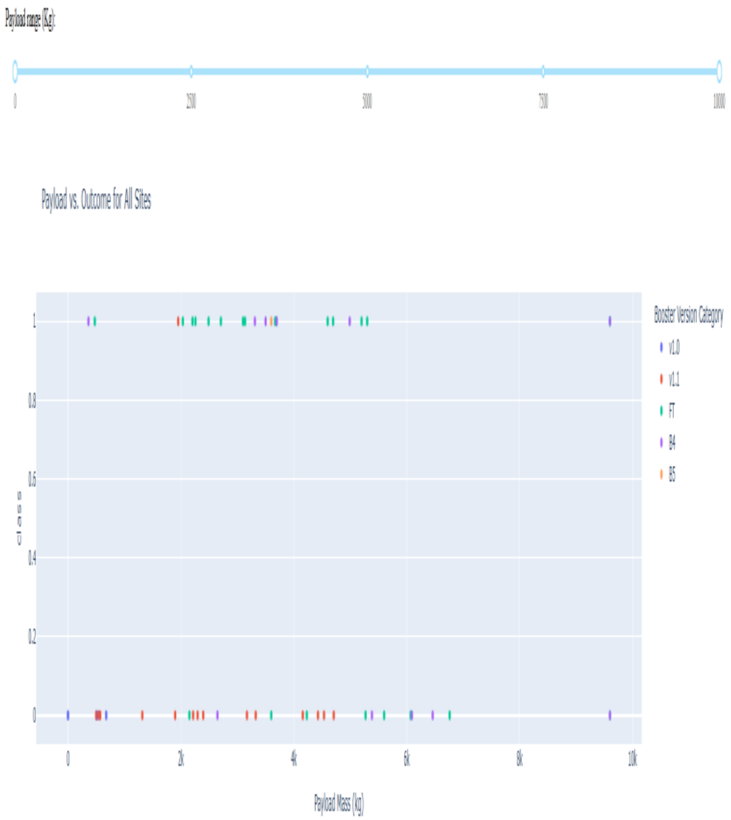
This section of the Plotly Dash dashboard allows users to explore how payload mass affects launch outcomes. It includes:

- Payload Range Slider: Users can filter launch data by selecting a specific payload mass range.
- Scatter Plot (Payload Mass vs. Launch Outcome): This plot visualizes the relationship between payload mass and launch success/failure. Each point represents a launch, color-coded by booster version.

Key features and interactions:

- Payload Filtering: The slider allows users to focus on specific payload ranges and observe the corresponding launch outcomes.
- Booster Version Comparison: Color-coding enables visual comparison of different booster versions' performance across various payload masses.
- Dynamic Updates: The scatter plot and its title update in real-time as users interact with the slider and other filters.

This interactive tool provides a dynamic way to understand how payload mass and booster version affect launch success.



Section 5

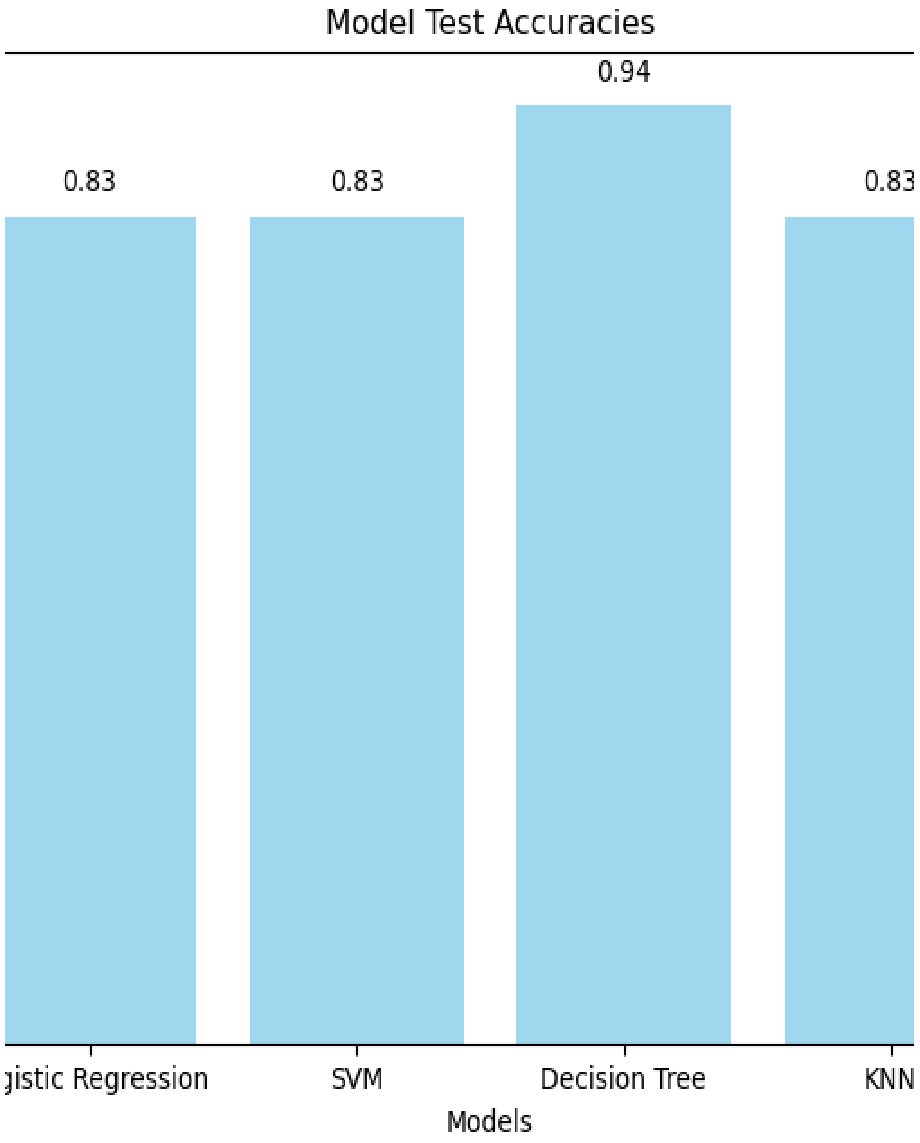
Predictive Analysis (Classification)

Classification Accuracy

This bar chart compares the test accuracies of four different classification models: Logistic Regression, SVM, Decision Tree, and KNN.

- Accuracy Scores: Each bar represents a model's accuracy, showing how well it predicts outcomes.
- Decision Tree Performance: The Decision Tree model has the highest accuracy (0.94), indicating strong predictive ability.
- Consistent Alternatives: Logistic Regression, SVM, and KNN achieve similar accuracies (0.83), suggesting reliable performance.
- Decision Tree Variability: While accurate, Decision Trees can be inconsistent across different runs due to their sensitivity to training data.

In conclusion, the Decision Tree model performs best in this test, but its variability should be considered. The other models offer consistent, albeit slightly less accurate, predictions. This information helps in choosing the right model based on the desired balance of accuracy and consistency.

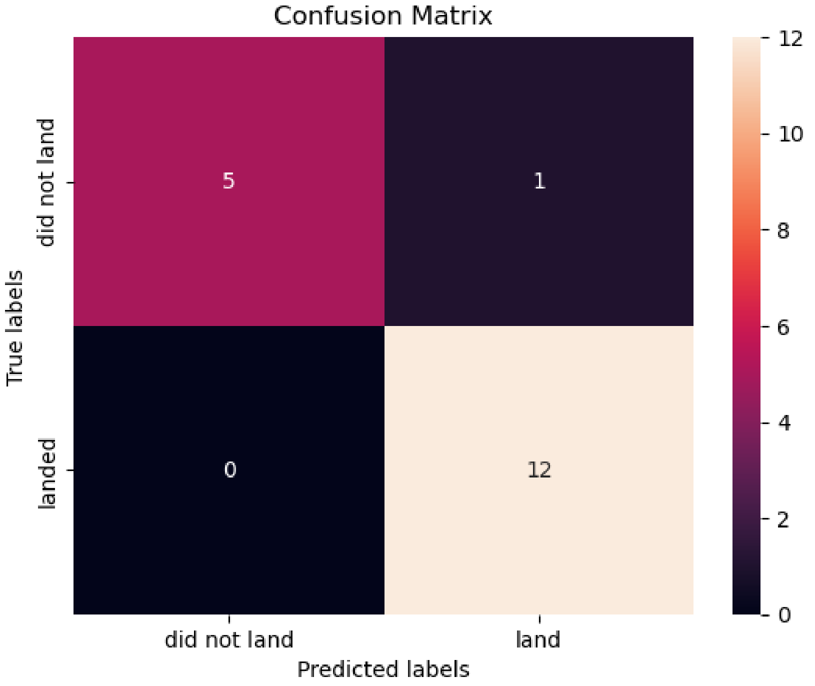


Confusion Matrix

This confusion matrix visualizes the performance of the Decision Tree model in predicting launch landing outcomes.

- True Outcomes: It compares the actual landing outcomes ("landed" or "did not land") with the model's predictions.
- Accurate Predictions:
  - 12 missions correctly predicted as "landed" (True Positives).
  - 5 missions correctly predicted as "did not land" (True Negatives).
- Misclassifications:
  - 1 mission incorrectly predicted as "landed" (False Positive).
  - 0 missions incorrectly predicted as "did not land" (False Negatives).
- Effective Landing Identification: The absence of False Negatives indicates the model's effectiveness in identifying successful landings.
- Decision Tree Variability: It's important to remember that Decision Tree performance can vary between runs due to its sensitivity to training data and its random tree-building process.

Overall, this confusion matrix shows that the Decision Tree model performed well in this specific run, particularly in identifying successful landings. However, potential variability should be considered when interpreting the results.



Conclusions

This project analyzed SpaceX launch data to identify success factors and predict landing outcomes. Key findings include:

- Launch Success Factors: Launch site, payload mass, and historical trends influence mission success.
- Geospatial Insights: Launch site locations and proximity to coastlines are important.
- Interactive Exploration: The Plotly Dash dashboard allows for dynamic data exploration.
- Predictive Accuracy: The Decision Tree model achieved the highest accuracy (86.25%) in predicting landing outcomes.

These insights can help Space Y:

- Optimize launch site selection and mission planning.
- Assess feasibility and risks of different missions.
- Develop competitive strategies based on SpaceX's performance.
- Understand the influence of geographical factors.
- Utilize the interactive dashboard and predictive models for decision-making.

This project demonstrates the value of data analysis and machine learning for gaining insights and informing strategic decisions in the space launch industry.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!